

RBG-48379

10 CFR 50.73

October 20, 2025

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2025-002-01, Unidentified Drywell Leakage
Increase Results in Shutdown

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report.

This document contains no commitments.

Should you have any questions, please contact Mr. Jack McCoy, Regulatory and Emergency Planning Manager, at 225-378-3310.

Respectfully,

**Jack
McCoy** 
Digitally signed by Jack McCoy
DN: cn=Jack McCoy, o=Manager
Regulatory and Emergency Preparedness,
ou=Entergy/River Bend,
email=jmccoy1@entergy.com
Date: 2025.10.20 13:34:06 -05'00'

JM/jdb

Enclosure: LER 50-458 / 2025-002-01, Unidentified Drywell Leakage
Increase Results in Shutdown

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station
NRC Office of Nuclear Security and Incident Response

Enclosure

RBG-48379

Licensee Event Report
50-458/2025-002-01

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.			EXPIRES: 04/30/2027			
 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)											
1. Facility Name River Bend Station, Unit 1					☒ 050 ☐ 052	2. Docket Number 00458		3. Page 1 of 3			
4. Title Unidentified Drywell Leakage Increase Results in Shutdown											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050 ☐ 052	Docket Number
05	21	2025	2025	002	01	10	20	2025	N/A		
9. Operating Mode 1					10. Power Level 100						
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☒ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)(i)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☒ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ Other (Specify here, in abstract, or in NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Jack McCoy, Regulatory and Emergency Planning Manager									Phone Number (Include Area Code) (225) 378-3310		
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable To IRIS		Cause	System	Component	Manufacturer	Reportable To IRIS	
E	BO	V	A585	Y							
14. Supplemental Report Expected						15. Expected Submission Date			Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date)									
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)											
On May 21, 2025, at 0324 CDT, with River Bend Station (RBS), Unit 1 operating in Mode 1 at 100% power, unidentified drywell leakage experienced a greater than 2 gallon per minute change over a 24-hour period. Technical Specifications (TS) Limiting Condition of Operations (LCO) 3.4.5 Condition C was entered. The required action to verify the source of unidentified leakage increase could not be completed within the allowable completion time. As a result, a plant shutdown was initiated at 0656 CDT. While investigating the unidentified leakage source, RBS determined that a degraded condition existed following the examination of the Residual Heat Removal (RHR) 'A' injection line testable check valve. The leakage from the RHR 'A' injection line testable check valve was determined to be Reactor Coolant System pressure boundary leakage. The direct cause of the leakage past RHR 'A' injection line testable check valve was due to improper gasket seating. This event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(A) as a Plant Shutdown Required by Technical Specifications, 10 CFR 50.73(a)(2)(ii)(A) as a Degraded or Unanalyzed Condition that resulted in the condition of the nuclear power plant, including its principal safety barriers being seriously degraded, and 10 CFR 50.73(a)(2)(i)(B) as an Operation or Condition Prohibited by Technical Specifications.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
River Bend Station, Unit 1	☒ 050	00458	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2025	002	01

NARRATIVE**Event Description**

On May 21, 2025, at 0324 CDT, with River Bend Station (RBS), Unit 1 operating in Mode 1 at 100% power, unidentified drywell leakage experienced a greater than 2 gallon per minute change over a 24-hour period. Technical Specifications (TS) Limiting Condition of Operations (LCO) 3.4.5 Condition C was entered. The required action to verify the source of unidentified leakage increase is not service sensitive type 304, type 316 austenitic stainless steel, or other inter-granular stress corrosion cracking susceptible material within 4 hours, could not be completed within the allowable completion time due to the drywell being inaccessible during plant operation. As a result, a controlled manual plant shutdown was initiated at 0656 CDT. At 1405 CDT, the control room operators placed the mode switch to Shutdown. The control room operators entered the appropriate post scram procedures and appropriately responded to the reactor scram.

This event was reported on May 21, 2025, at 0950 CDT in accordance with 10 CFR 50.72(b)(2)(i), as an event or condition that results in a TS required plant shutdown. (EN 57725)

During the investigation of the unidentified leakage following the shutdown, RBS determined that a degraded condition existed following the performance of a visual examination of E12-AOVF041A (Residual Heat Removal 'A' Injection Line Testable Check Valve) [BO:V]. This examination was conducted on May 22, 2025, at 2358 CDT.

This event was reported on May 23, 2025, at 0121 CDT in accordance with 10 CFR 50.72(a)(2)(ii)(A), as Degraded or Unanalyzed Condition that resulted in the condition of the nuclear power plant, including its principal safety barriers being seriously degraded. (EN 57731)

The elevated drywell leakage was determined to be Reactor Coolant System (RCS) [AB] pressure boundary leakage and required actions were not completed within the completion time as required by TS LCO 3.4.5. Therefore, this event is also being reported under 10 CFR 50.73(a)(2)(i)(B) as an Operation or Condition Prohibited by TS.

Event Cause

The direct cause of this event was improper gasket seating, resulting from erosion of the gasket seating surface. The erosion was caused by steam cutting to the base metal of the gasket seat surface. In this degraded state, the base metal did not provide the required support as designed. Two causal factors were identified; (1) The organization did not fully understand the degraded condition of E12-AOVF041A valve body due to steam cutting; therefore, an adequate repair strategy was not formulated or executed and (2) Required VT-3 inspections were not added to the Job Plan as the plan was in Approved status and therefore not routed for additional reviews. In addition, the inspectors failed to realize the need for an inspection as it was not listed in the Job Plan description.

Safety Assessment

The actual consequence of these events was the initiation of a controlled manual plant shutdown. Following the controlled manual plant shutdown, reactor pressure was maintained by the Turbine Steam Drains [TF] and reactor water level was maintained by the Feedwater [SJ] system.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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	☐ 052		2025	002	01

Corrective Actions

In Progress:

- IT to resolve the discrepancies identified within the action tracking program.

Completed:

- Completed a work order to repair and restore E12-AOVF041A to pre-steam cut conditions.
- Revise job plans to include Operational Experience from this event.
- Reinforce requirements of ASME Section XI inspections required for Class 1 pumps and valves with engineering.

Previous Occurrences:

None

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX]. River Bend equipment codes are identified as (XX).